

Derrin Bright

Bioinformatics | Biotechnology | ML in Biology

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Education

National University of Singapore (NUS)

Master of Science in Precision Health and Medicine, Yong Loo Lin School of Medicine

Vellore Institute of Technology (VIT)

Bachelor of Technology in Biotechnology

Singapore

Aug 2026-Present

CGPA: 8.8 Vellore, India

2022-2026

Technical Skills

Computational Biology Tools	BLAST AlphaFold AutoDock Vina GROMACS MGLTools Bioconductor OpenBabel Cytoscape Seurat GSEA FIMO DESeq2 WGCNA MSigDB limma
Data Analysis Tools	Neo4J KNIME NetworkAnalyst TCGA/GDC
Programming Tools	R Python
Machine Learning Tools	TensorFlow Keras NumPy Pandas Matplotlib scikit-learn XGBoost
Certificates	Machine Learning A-Z Python for Data Science, AI & Development Bulk RNA-Seq Data Analysis Advanced Bioinformatics Single-Cell RNA-Seq Data Analysis using R & Python

Research Experiences

Project Intern at Nanyang Technological University, Singapore

Jan 2026 - June 2026

- Built a transcription factor activity inference pipeline on large-scale postmortem brain RNA-seq data, using VIPER to infer regulon activity and linear mixed-effects models to test associations with neurodegenerative disease progression.
- Extended the analysis from bulk tissue to single-nucleus RNA-seq, modelling transcription factor activity separately across neuronal and glial cell types to test whether the pathology associations were specific to particular cell types.
- Supervised by Dr. Guillaume Thibault | Guillaume Thibault Lab

Research Intern at Boston University, USA

Sept 2025 - April 2026

- Scanned Long Control Region sequences across 207 HPV types and identified 104 transcription factors with significantly different binding site densities between high-risk and low-risk types.
- Integrated results with TCGA transcriptomics and public ChIP-seq data, finding over half of the high-risk-enriched TFs expressed in HPV-positive tumors and confirming LCR occupancy for seven of them.
- Analyzed sequence conservation across HPV-16 and HPV-18 clinical isolates and found the identified binding sites were highly conserved across strains.
- First author on the resulting paper in *Viruses* (2026).
- Supervised by Dr. Juan Fuxman Bass

Research Intern at University of Liverpool, UK

Sept 2025 - August 2026

- Built and released a catalogue of 30 genome-scale metabolic models for five immune cell types across CLL, Richter Transformation and healthy donors, reconstructed from a large single-cell bootstrap ensemble using ftINIT and GECKO 3 enzyme constraints.
- Compared the transformed state against both CLL and a matched healthy baseline to separate metabolic capabilities that were genuinely new from ones that were simply restored.
- Ran flux balance analysis and flux variability analysis across the model catalogue to characterise metabolic capability and flux behaviour for each cell type and condition.
- Supervised by Dr. Francesco Del Carratore

Research Intern at IIT Bombay, India

May 2025 - Sept 2026

- Performed differential expression analysis and WGCNA across four autoimmune diseases (SLE, rheumatoid arthritis, type 1 diabetes, ankylosing spondylitis) to identify disease-associated co-expression modules and hub genes.
- Followed up with Gene Set Enrichment Analysis and PPI network construction using NetworkAnalyst to uncover dysregulated pathways and functional gene relationships shared across the four diseases.
- Built and evaluated classification models (Logistic Regression, SVM, XGBoost and a deep neural network) to distinguish disease from healthy states based on gene expression, and tested how well the models generalized across the four diseases.
- Supervised by Dr. Prakriti Tayalia

Research Intern at Ashoka University, India

Jan 2025 - Mar 2025

- Analyzed five assembled *E. coli* genomes for carbapenem and beta-lactam resistance using BLAST and RGI-CARD to identify mutations, and correlated genotype-phenotype relationships through AST report analysis.
- Performed Multiple Sequence Alignment (MSA) of Penicillin-Binding Proteins (PBPs) from the five *E. coli* genomes against a reference wild-type sequence to detect conserved mutations, and conducted genomic analysis that led to the identification of a 12-base pair insertion in the PBP3 gene, contributing to antibiotic resistance mechanisms.
- Automated genomic data processing by developing a custom script to systematically rename and organize FASTA files, creating a structured dataset for phylogenetic analysis using the PhaME tool.

- Supervised by Dr. Shraddha Karve

Biotech Team Lead at NovaLinks, India

Feb 2025 - April 2025

- Led a project integrating miRNA analysis and histopathological imaging with machine learning to enhance lung cancer subtype classification and improve diagnostic precision.
- Conducted research on miRNA biomarkers specific to lung cancer and provided biological insights to support the development of an accurate classification model for distinguishing adenocarcinoma from squamous cell carcinoma.

Publications

Comparative Analysis of Transcription Factor Binding Sites in the Long Control Region Across Human Papillomavirus Types | *Viruses*, 2026, 18(6):646 (MDPI)

Integrative transcriptomic analysis identifies predictive gene signatures across autoimmune diseases: implications for personalized therapeutic targeting | *Under review at Journal of Autoimmunity (Elsevier)*

HPV-Associated Transcriptional Landscape in Cervical and Head and Neck Cancers | *In revision at Infectious Agents and Cancer (Springer Nature)*

Beyond the HPV Imprint: Cervicovaginal Dysbiosis and Microbiome-Mediated Modulation of HPV Pathogenesis and Cervical Cancer Progression | *Submitted to Scientifica (Hindawi)*

Technical Projects

HPV-Associated Transcriptional Landscape in Cervical and Head and Neck Cancers

- Integrated differential expression analysis and WGCNA across TCGA-CESC and TCGA-HNSC cohorts, loaded via TCGAbiolinks and batch-corrected with ComBat-seq, to identify 52 transcriptional alterations shared between HPV-positive cervical and head and neck cancers.
- Used an elastic-net and mRMR feature-selection pipeline with Borda rank aggregation to prioritise a six-gene panel (CDKN2A, EFNB3, CDKN2B-AS1, COL28A1, MCM2, KCNS1) that consistently separated HPV-positive from HPV-negative tumours, and validated it in two independent GEO cohorts.
- Validated the panel at single-cell resolution using Seurat and Harmony integration across two additional scRNA-seq datasets, finding the six genes predominantly expressed in epithelial cells, and mapped their regulatory transcription factors and miRNAs using NetworkAnalyst and Cytoscape.

Computational Identification of Antimalarial Leads through Fingerprint Similarity Analysis

[GitHub Link](#)

- Screened a library of antimalarial molecules against a query (chloroquine drug) by generating molecular fingerprints and computing Tanimoto similarity coefficients using KNIME and RDKit nodes.
- Extracted molecular data from the Protein Data Bank (PDB) and visualized pairwise similarity scores as a heatmap to identify high-similarity candidates for antimalarial drug development.

In Silico Analysis of 3DTC/CEP-6331 Complex using Molecular Docking

[GitHub Link](#)

- Implemented a molecular docking simulation using AutoDock Vina to model the interaction between the 3DTC kinase and the CEP-6331 inhibitor, achieving a predicted binding affinity of -12.23 kcal/mol.
- Analyzed the top binding pose to identify the specific amino acids responsible for the stable interaction, including key hydrophobic (ILE180, LEU274) and polar (LYS228, ASP284) residues.

Single-Cell Transcriptomic Profiling of the B16 Melanoma's Microenvironment

[GitHub Link](#)

- Processed raw single-cell transcriptomic data from a B16 melanoma model using R (Seurat), applying quality control, normalization, and clustering to map the tumor's cellular heterogeneity.
- Characterized the resulting cell clusters by annotating fourteen distinct cell types within the tumor microenvironment and performed differential expression analysis to identify their unique gene signatures.

Honors and Leadership

First runner-up in the Biomimicry Innovation Challenge and Showcase, Vellore Institute of Technology, 2024

Top 10 Winner of Bio-Inspired Design Fest (BIDFEST) Ideathon, Vellore Institute of Technology, 2024

Vice Chairperson at Alpha Bio Cell, VIT University, India

January 2024 - December 2024

- Led several biotechnology and bioinformatics-related events, including hackathons, workshops, and quizzes, enhancing the club's engagement.
- Mentored junior core members and applied leadership, technical, and project management skills to ensure successful event planning and execution.

Extra-Curricular

Volunteer | National Service Scheme, NGO

Conducted sessions in government schools on career opportunities and the importance of joining the armed forces.

Volunteer | Becoming I Foundation, NGO

Taught science, math, and english in government schools, supporting underprivileged communities.

Event Coordinator | BioSummit, School of Biosciences and Technology

Organized a flagship event connecting academia and food industries, managing food distribution and ensuring smooth execution.